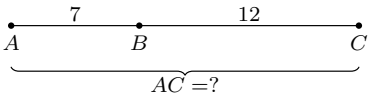
Geometry Review: Unit 1 — Basics & Reasoning

Station 1 of 4: Segments, Midpoints, and Distance

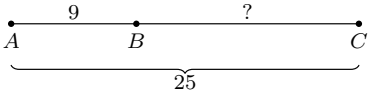
Name: _____ Date: _____

Show your work. Give exact answers; round decimals to the nearest tenth only when asked. Diagrams are not drawn to scale.

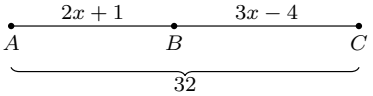
1. Points A, B, and C are collinear with B between A and C. If $AB = 7$ and $BC = 12$, find AC.



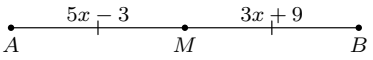
2. Points A, B, and C are collinear with B between A and C. If $AC = 25$ and $AB = 9$, find BC.



3. B is between A and C. If $AB = 2x + 1$, $BC = 3x - 4$, and $AC = 32$, find x, AB, and BC.



4. M is the midpoint of $\overline{AB}$. If $AM = 5x - 3$ and $MB = 3x + 9$, find x and AB.



5. Find the midpoint of the segment with endpoints $A(2, -4)$ and $B(8, 6)$.

6. $M(1, 3)$ is the midpoint of $\overline{AB}$. If A is $(-3, -1)$, find the coordinates of B.

7. Find the distance between $(1, 2)$ and $(7, 10)$.

8. Find the distance between $(-2, 3)$ and $(4, 6)$. Give the exact answer and a decimal rounded to the nearest tenth.

9. On a number line, P is at -3 and Q is at 9. Find PQ and the coordinate of the midpoint of $\overline{PQ}$.

10. Point Z lies on $\overline{XY}$ with $XY = 30$ and $XZ : ZY = 2 : 3$. Find XZ and ZY.

Geometry Review: Unit 1 — Basics & Reasoning

Station 2 of 4: Angle Relationships and Angle Bisectors

Name: _____ Date: _____

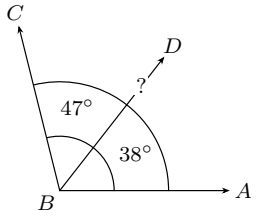
Show your work. Include degree symbols in your answers. Diagrams are not drawn to scale.

1. Classify each angle as acute, right, obtuse, or straight: 35°, 90°, 140°, 180°.

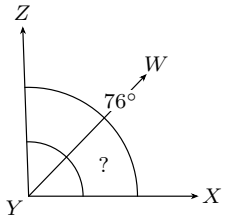
2. Find the complement of 62°.

3. Find the supplement of 115°.

4. $m\angle ABD = 38°$ and $m\angle DBC = 47°$. Find $m\angle ABC$.



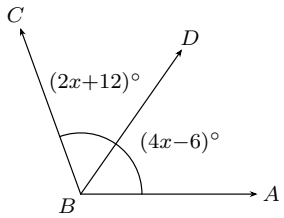
5. $\overrightarrow{YW}$ bisects $\angle XYZ$, and $m\angle XYZ = 76°$. Find $m\angle XYW$.



6. Two complementary angles measure $x°$ and $(2x + 15)°$. Find both angles.

7. Two supplementary angles measure $(4x + 10)°$ and $(6x - 20)°$. Find both angles.

8. $\overrightarrow{BD}$ bisects $\angle ABC$. $m\angle ABD = (4x - 6)°$ and $m\angle DBC = (2x + 12)°$. Find $m\angle ABC$.



9. What is the measure of the angle formed by the hands of a clock at 3:00?

10. An angle's supplement is 30° more than twice its complement. Find the angle.

Geometry Review: Unit 1 — Basics & Reasoning

Station 3 of 4: Conditional Statements and Their Related Forms

Name: _____ Date: _____

Answer each question about the statements.

1. Identify the hypothesis and the conclusion: "If it rains, then the ground is wet."

2. Write the converse of: "If a number is divisible by 4, then it is even."

3. Is the converse in problem 2 true or false? If false, give a counterexample.

4. Write the inverse of: "If $x > 5$, then $x > 3$."

5. Write the contrapositive of: "If $x > 5$, then $x > 3$."

6. Is the statement "If $x > 5$, then $x > 3$" true or false? Is its contrapositive true or false?

7. Give a counterexample to: "If a number is prime, then it is odd."

8. Rewrite as an if-then statement: "All squares have four sides."

9. Combine "If an angle measures $90°$, then it is a right angle" and its converse into one biconditional statement.

10. The statement "If a triangle is equilateral, then it is isosceles" is true. Write its contrapositive.

Geometry Review: Unit 1 — Basics & Reasoning

Station 4 of 4: Inductive and Deductive Reasoning

Name: _____ Date: _____

Find patterns, make conjectures, and apply logic. Show your thinking.

1. Find the next two numbers: $3, 6, 12, 24, \ldots$

2. Find the next two numbers: $2, 5, 8, 11, \ldots$

3. Find the next two numbers: $1, 4, 9, 16, 25, \ldots$

4. Find the next two numbers: $81, 27, 9, 3, \ldots$

5. Write a rule for the nth term of $4, 7, 10, 13, \ldots$ and find the 50th term.

6. Is each an example of inductive or deductive reasoning? (a) "Every sunset I have seen was in the west, so the sun always sets in the west." (b) "All mammals are warm-blooded. A whale is a mammal. So a whale is warm-blooded."

7. Use the Law of Detachment. Given: "If you study, then you will pass." Mia studied. What can you conclude?

8. Use the Law of Syllogism. Given: "If it snows, school is closed" and "If school is closed, we sleep in." What can you conclude?

9. Give a counterexample: "The sum of two numbers is always greater than either number."

10. Segments connect every pair of n points. For $2, 3, 4, 5$ points there are $1, 3, 6, 10$ segments. Predict the number for 6 points, and write a rule in terms of n.

Geometry Review: Unit 1 — Basics & Reasoning

Answer Key

Station 1: Segments, Midpoints, and Distance

1. $AC = 19$
2. $BC = 16$
3. $x = 7,\ AB = 15,\ BC = 17$ $\quad 5x - 3 = 32$
4. $x = 6,\ AB = 54$ $\quad 5x - 3 = 3x + 9;\ AM = 27$
5. $(5, 1)$
6. $B(5, 7)$
7. 10 $\quad \sqrt{6^2 + 8^2}$
8. $3\sqrt{5} \approx 6.7$ $\quad \sqrt{36 + 9} = \sqrt{45}$
9. $PQ = 12;$ midpoint $= 3$
10. $XZ = 12,\ ZY = 18$

Station 2: Angle Relationships and Angle Bisectors

1. acute, right, obtuse, straight
2. $28°$
3. $65°$
4. $85°$
5. $38°$
6. $25°,\ 65°$ $\quad 3x + 15 = 90$
7. $86°,\ 94°$ $\quad 10x - 10 = 180,\ x = 19$
8. $60°$ $\quad 4x - 6 = 2x + 12,\ x = 9$
9. $90°$
10. $30°$ $\quad 180 - x = 2(90 - x) + 30$

Station 3: Conditional Statements and Their Related Forms

1. Hypothesis: it rains. Conclusion: the ground is wet.
2. If a number is even, then it is divisible by 4.
3. False; counterexample: 6 is even but not divisible by 4
4. If $x \leq 5$, then $x \leq 3$.
5. If $x \leq 3$, then $x \leq 5$.
6. Both are true (a conditional and its contrapositive always have the same truth value)
7. 2 is prime but even
8. If a figure is a square, then it has four sides.
9. An angle measures $90°$ if and only if it is a right angle.
10. If a triangle is not isosceles, then it is not equilateral.

Station 4: Inductive and Deductive Reasoning

1. $48,\ 96$ $\quad$ *multiply by 2*
2. $14,\ 17$ $\quad$ *add 3*
3. $36,\ 49$ $\quad$ *perfect squares*
4. $1,\ \dfrac{1}{3}$ $\quad$ *divide by 3*
5. $3n + 1;\ 151$
6. Inductive; deductive
7. Mia will pass.
8. If it snows, then we sleep in.
9. $5 + (-2) = 3$, which is not greater than 5
10. $15;\ \dfrac{n(n-1)}{2}$